

Exam

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Electricity and Electronics EBN 111

20 July 2020

Test Information

Maximum marks:	96	Full marks:	90
Duration of test:	240 min	Open/Closed book:	Open
Total number of pages (this page included)		30	

Important

1. The examination regulations of the University of Pretoria apply.
2. This test is completed on ClickUP.
3. Answers must include units where applicable.
4. Final answers must be written as real numbers and not as fractions.
5. Use at least 3 significant numbers to write irrational numbers.
6. If a question is asking for the real part or the imaginary part of a complex number, do not give j or the unit in the answer.

Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye

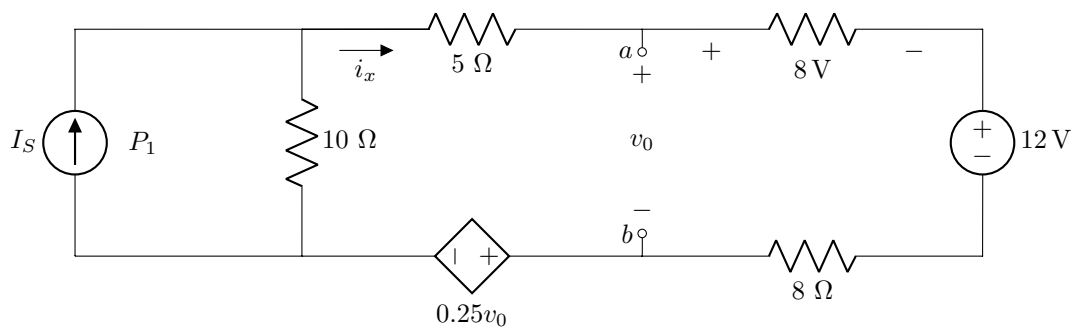
Section 1 / Afdeling 1

Question 1 [2]

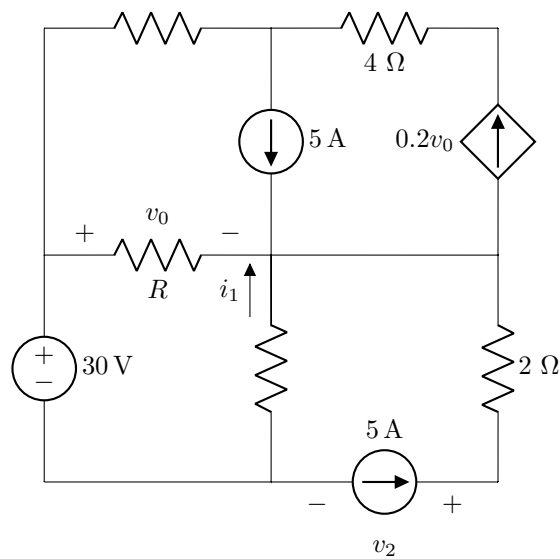
- | | |
|-----------|--|
| 01 | <p>A 9 V battery is used to power a remote-controlled car. The power delivered by the battery moves linearly from α W at $t = 0$ min to β W at $t = 100$ min. Determine the total energy used by the car.</p> <p>(The values of α and β are given in ClickUP.)</p> |
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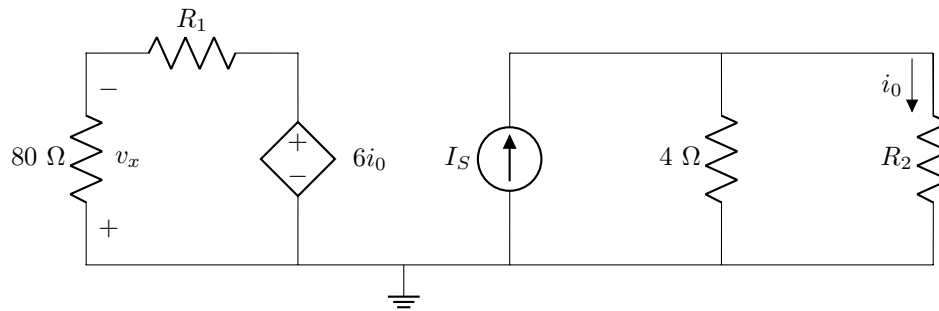
Questions 2 to 3 [4]

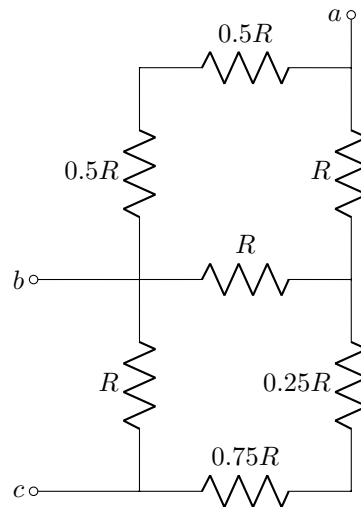
(The values of I_S and i_x are given in ClickUP.)



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| 02 | Calculate the power associated with the component labelled as P_1 . [2] |
| 03 | Determine v_0 . [2] |

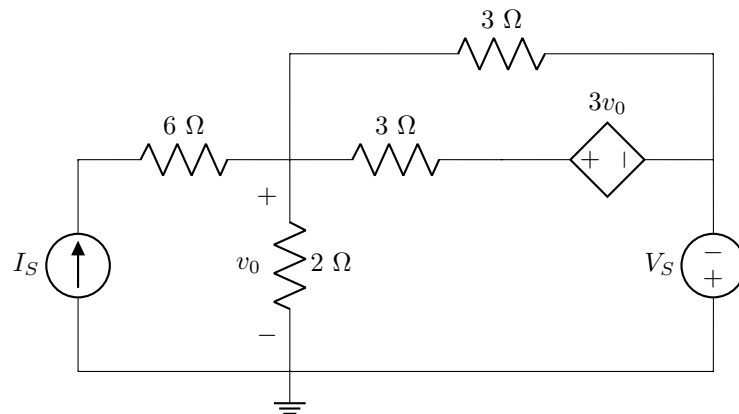
Questions 4 to 5 [4]Determine i_1 and v_2 .(The values of v_0 and R are given in ClickUP.)**04** i_1 [2]**05** v_2 [2]

Questions 6 to 7 [4]Determine I_S and v_x .(The values of i_0 , R_1 and R_2 are given in ClickUP.)**06** I_S [2]**07** v_x [2]

Questions 8 to 9 [4](The value of R is given in ClickUP.)**08** Determine R_{bc} . [2]**09** Determine R_{ac} . [2]

Question 10 [2]

(The values of V_S and I_S are given in ClickUP.)



10 Determine the voltage v_0 .

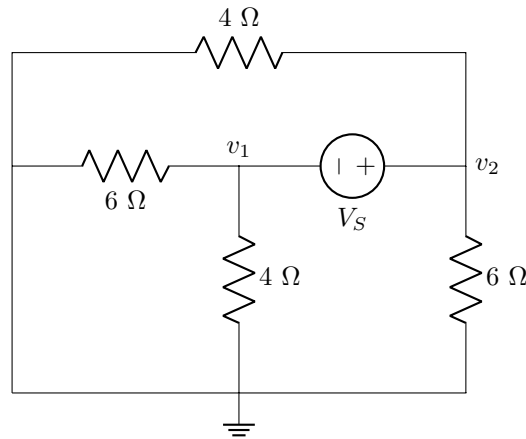
Question 11 [2]

Nodal analysis can be used to solve for the two node voltages shown below. The circuit contains an example of a supernode. The KVL equation for the supernode is as follows:

$$-v_1 + v_2 = V_S$$

Set up the KCL equation for the supernode and solve for v_2 .

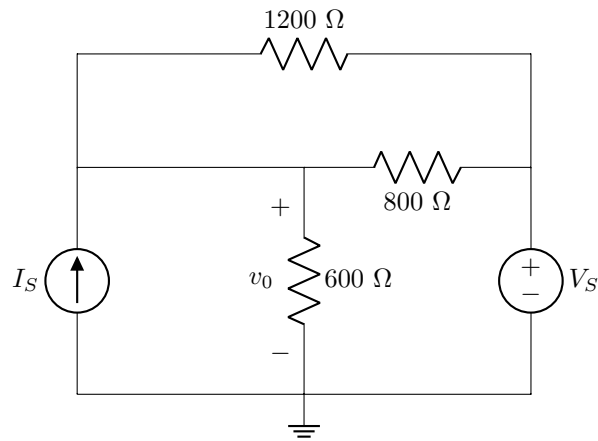
(The value of V_S is given in ClickUP.)



11 | v_2 [2]

Question 12 [2]

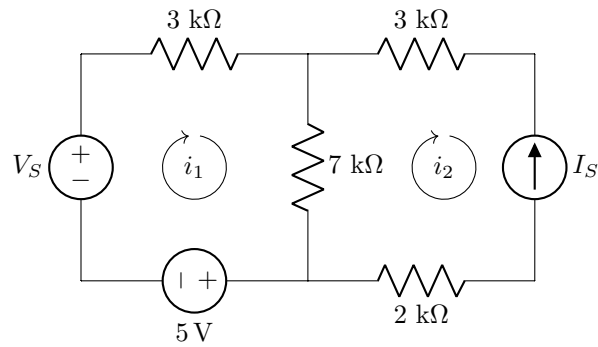
(The values of V_S and I_S are given in ClickUP).



12 Determine v_0 [2]

Question 13 [2]

(The values of V_S and I_S are given in ClickUP).



13 Determine i_1 [2]

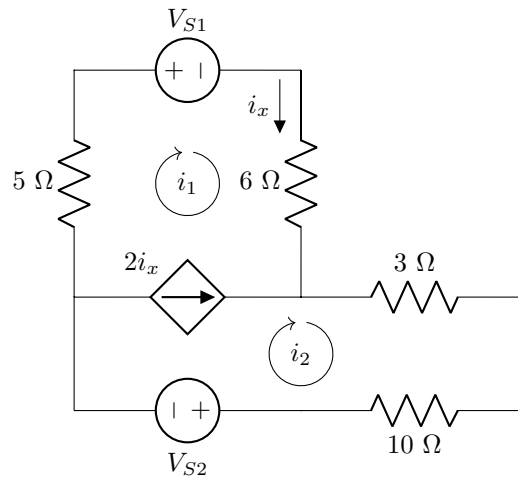
Question 14 [2]

The circuit below contains an example of a supermesh. The following is known:

$$-i_1 + i_2 = 2i_x \quad (\text{KCL equation for dependent current source } 2i_x)$$

Using this information, set up the KVL supermesh equation and solve for i_x .

(The values of V_{S1} and V_{S2} are given in ClickUP).

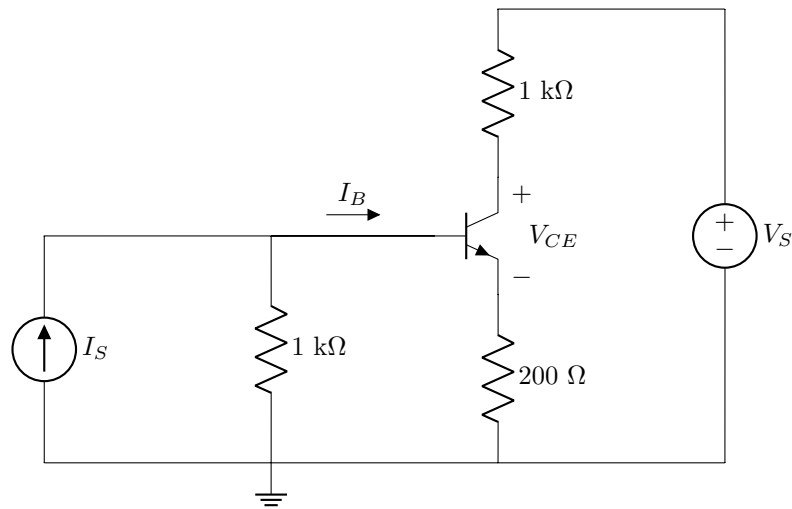


14 i_x [2]

Questions 15 to 16 [4]

It is known that $\beta = 99$ and $V_{BE} = 0.7$ V. Determine I_S and V_{CE} .

(The values of I_B and V_S are given in ClickUP.)



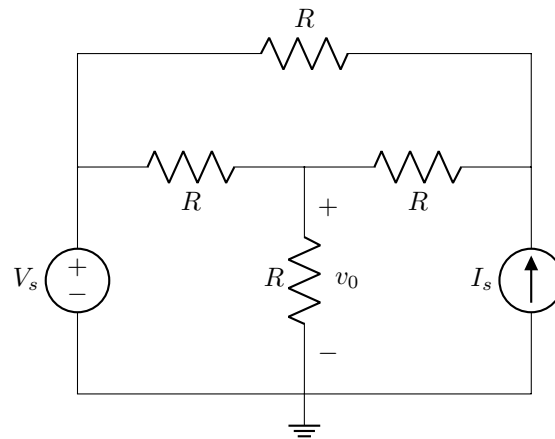
15 I_S [2]

16 V_{CE} [2]

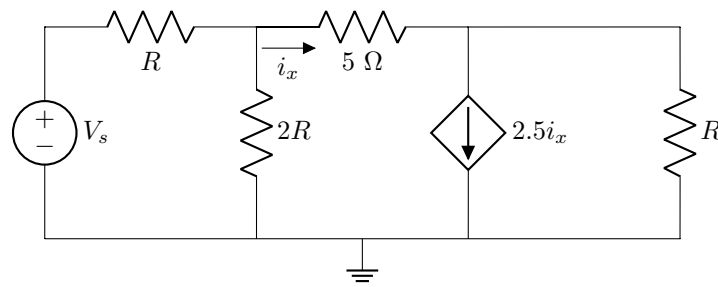
Total Section 1 / Totaal Afdeling 1: [32]

Section 2 / Afdeling 2

Questions 17 to 18 [4]



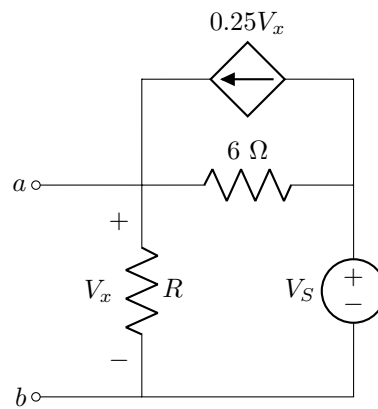
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| 17 | Determine the contribution to v_0 of the voltage source V_s alone if $I_s = 1$ mA and $R = 6$ k Ω . [2]
(The value of V_s is given in ClickUP.) |
| 18 | Determine the contribution to v_0 of the current source I_s alone if $V_s = 10$ V. [2]
(The values of I_s and R are given in ClickUP.) |

Question 19 [2]

- 19** Use source transformation to determine i_x . [2]
(The values of V_s and R are given in ClickUP.)

Questions 20 to 21 [4]

Find the Thevenin equivalent voltage V_{TH} and Thevenin equivalent resistance R_{TH} , with reference to terminals $a - b$.
(The values of V_S and R are given in ClickUP.)



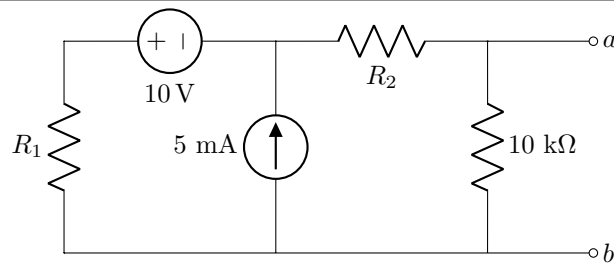
20 Determine V_{TH} . [2]

21 Determine R_{TH} . [2]

Questions 22 to 23 [4]

Find the Norton equivalent current I_N and Norton equivalent resistance R_N , with reference to terminals $a - b$.

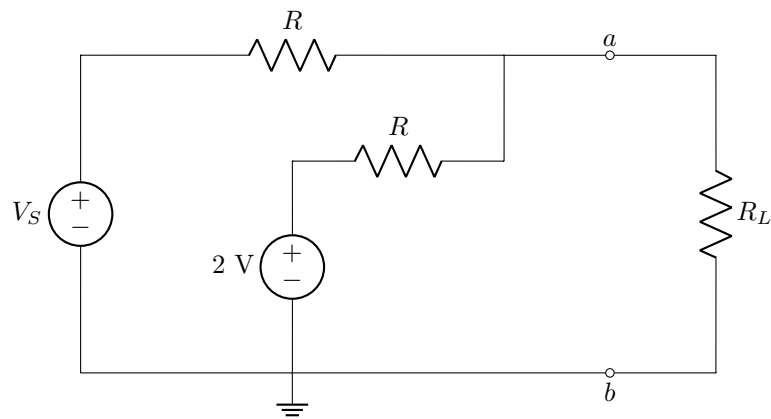
(The values of R_1 and R_2 are given in ClickUP.)



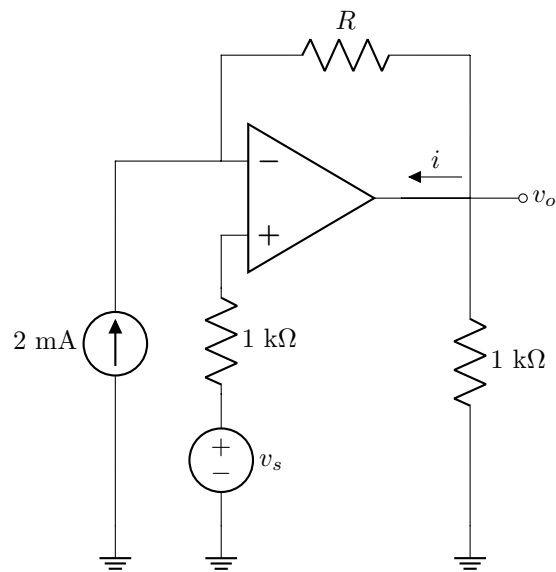
22 Determine I_N . [2]

23 Determine R_N . [2]

Questions 24 to 25 [4]

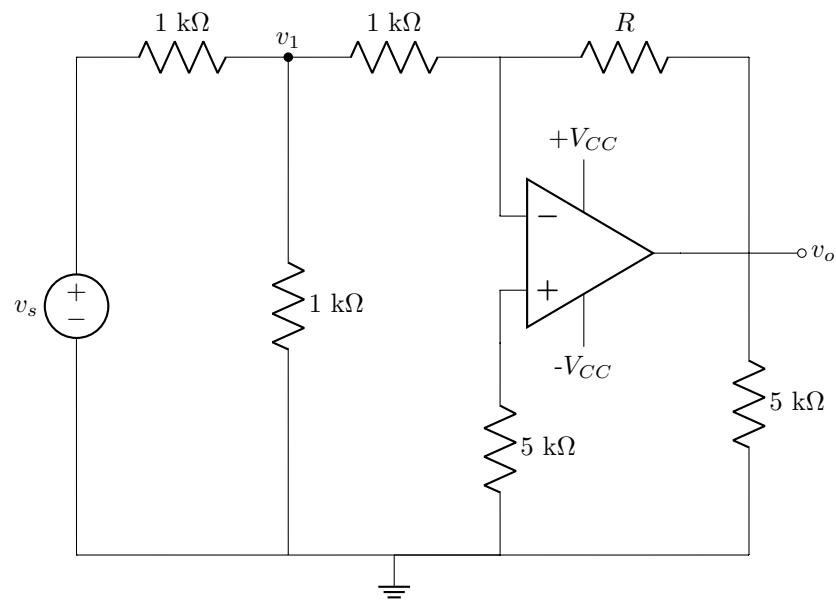


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| 24 | Determine the voltage V_L across the load R_L for maximum power transfer if $R = 250\ \Omega$. [2]
(The value of V_S is given in ClickUP.) |
| 25 | Determine the value of R such that the maximum power P_{max} is delivered to the load, given that $V_s = 2\text{ V}$ and $V_L = 1\text{ V}$. [2]
(The value of P_{max} is given in ClickUP.) |

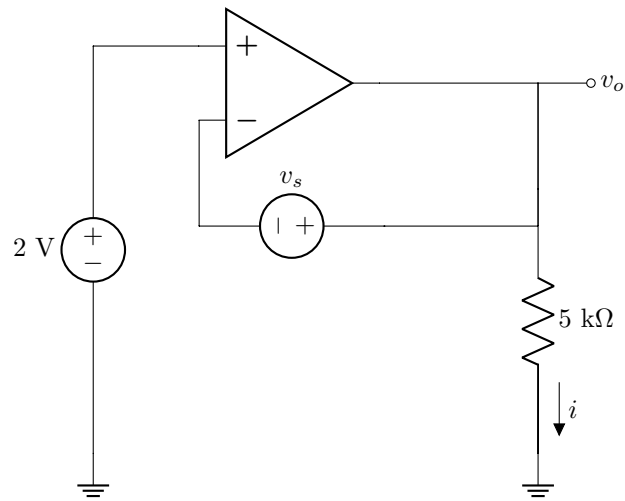
Question 26 [2]

- 26** Find i . [2]
(The values of v_s and R are given in ClickUP.)

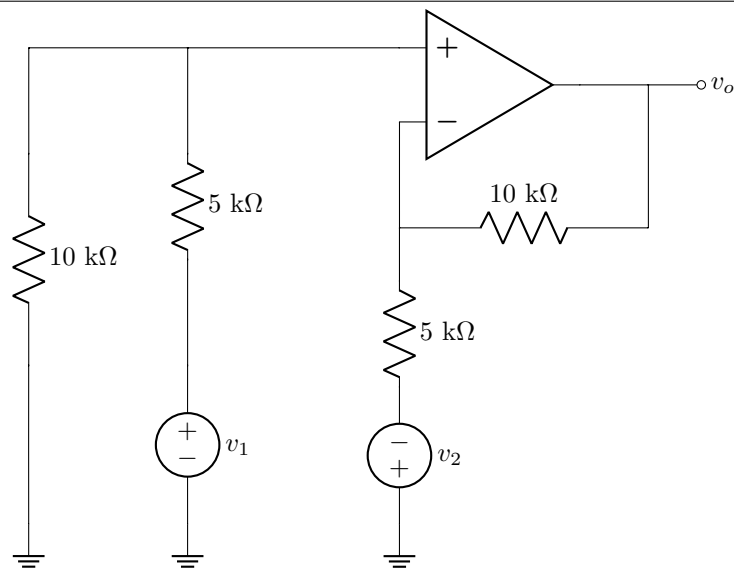
Questions 27 to 28 [4]



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| 27 | Determine v_o . [2]
(The values of v_s and R are given in ClickUP.) |
| 28 | Determine the maximum value of v_1 before the op-amp goes into saturation. [2]
(The values of V_{CC} and R are given in ClickUP.) |

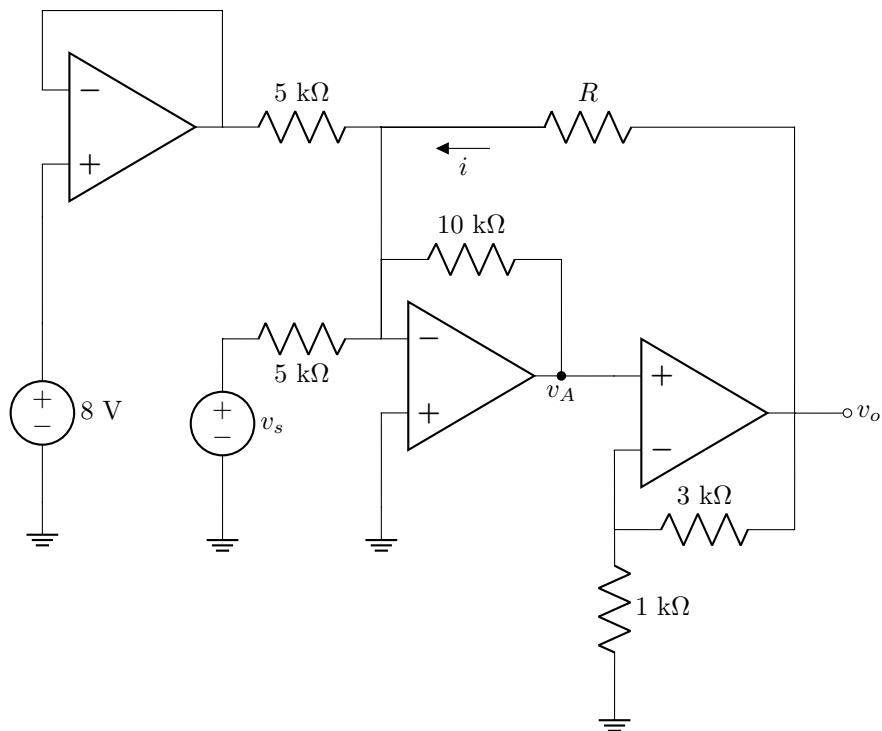
Question 29 [2]

- 29** Find i . [2]
(The value of v_s is given in ClickUP.)

Question 30 [2]

- 30** Find v_o . [2]
(The values of v_1 and v_2 are given in ClickUP.)

Questions 31 to 32 [4]

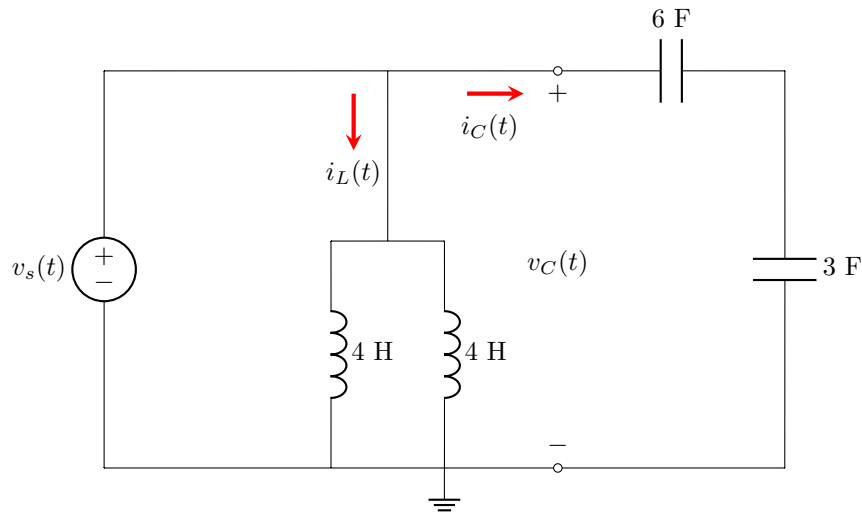


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| 31 | Determine v_A when $R \rightarrow \infty$. [2]
(The value of v_s is given in ClickUP.) |
| 32 | Determine i when $v_s = 6$ V. [2]
(The values of R and v_A are given in ClickUP.) |

Total Section 2 / Totaal Afdeling 2: [32]

Section 3 / Afdeling 3**Questions 33 to 35 [6]**

In the circuit below, $v_s(t) = \alpha e^{-0.2t}$ V and $i_L(0) = 2$ A.
The value of α is given in ClickUP.



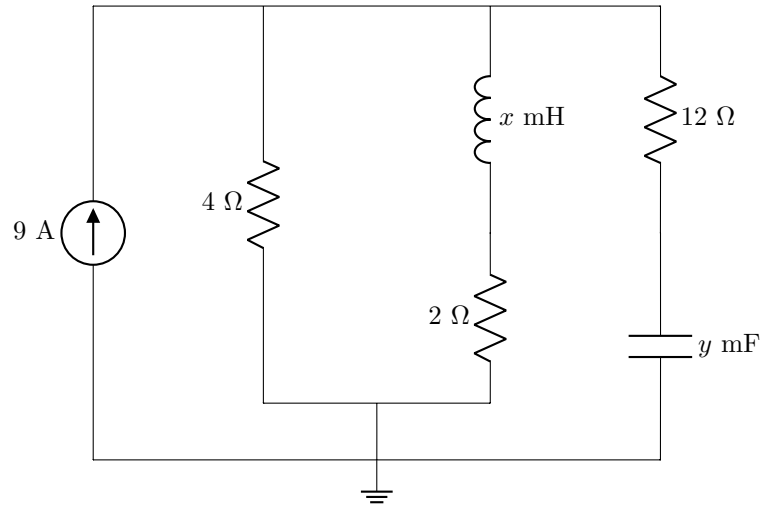
33 Determine $v_C(0)$. [2]

34 Determine $i_C(5)$. [2]

35 Determine $i_L(5)$. [2]

Questions 36 to 37 [4]

The circuit below operates under steady-state DC conditions.
The values x and y are given in ClickUP.



36 Determine the energy stored in the inductor w_L . [2]

37 Determine the energy stored in the capacitor w_C . [2]

Question 38 [2]

Sinusoids $v_1(t)$ and $v_2(t)$ have the following general form: $A \cos(\omega t + \phi)$.

It is known that:

- v_2 leads v_1 by σ° .
- v_2 lags v_3 by 100° .

The value of σ is given in ClickUP.

38 What is the phase angle θ between v_1 and v_3 such that $0 \leq \theta \leq 180^\circ$? [2]

Question 39 [2]

A current source of $i(t) = 2 \cos(100t + 90^\circ)$ A is connected in parallel with a resistor of $R \Omega$, a capacitor of $C \mu\text{F}$, and an inductor of $L \text{ mH}$. (In other words, all four elements are in parallel.) Determine the phasor voltage $\mathbf{V}$ across the current source, assuming the sign of the voltage is positive. Values of R , C , and L are given in ClickUP.

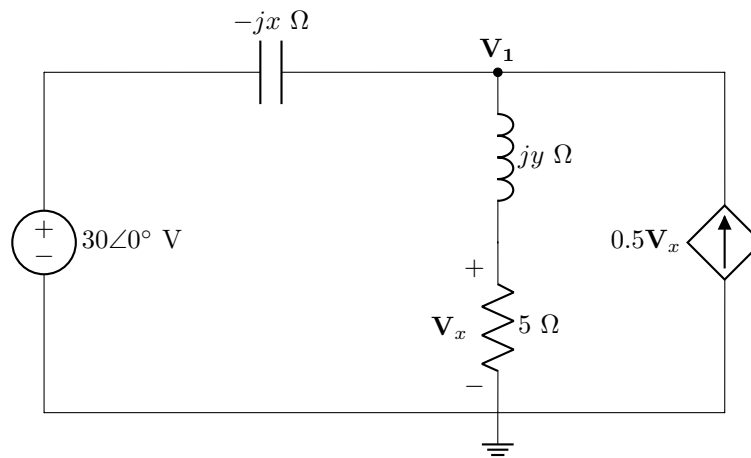
39 (a) If $\mathbf{V} = A + jB$, what is the real part A ? [2]

39 (b) If $\mathbf{V} = A + jB$, what is the imaginary part B ? [2]

Question 40 [2]

Find $\mathbf{V}_1$ in the circuit below.

The values x and y are given in ClickUP.

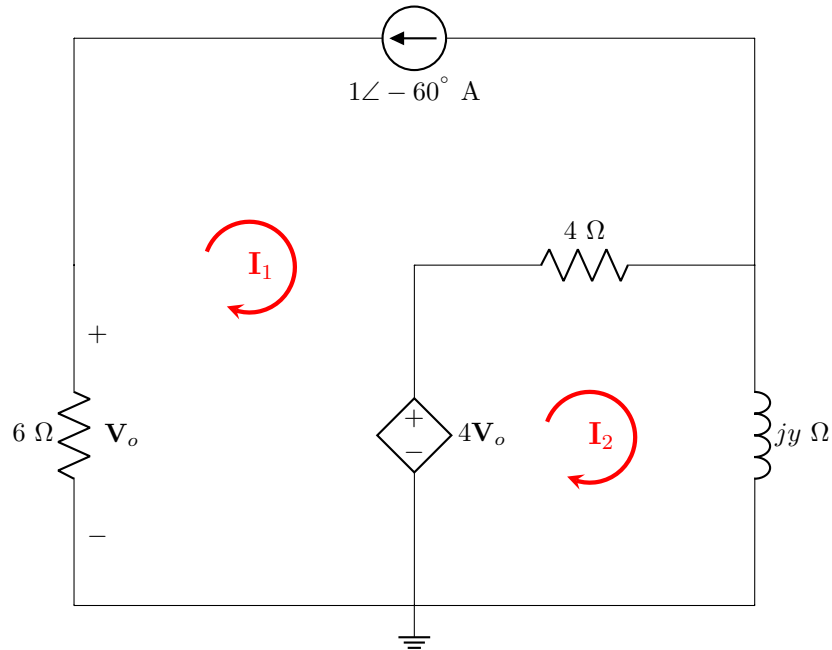


40 (a) If $\mathbf{V}_1 = C + jD$, what is the real part C ? [2]

40 (b) If $\mathbf{V}_1 = C + jD$, what is the imaginary part D ? [2]

Question 41 [2]

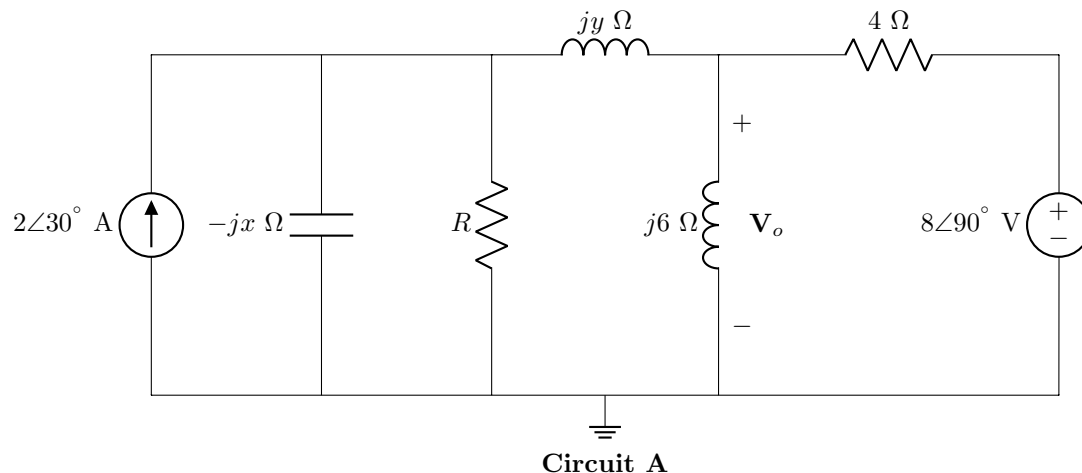
In the circuit below, use **mesh** analysis to find $\mathbf{I}_2$.
The value of y is given in ClickUP.



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| 41 | (a) If $\mathbf{I}_2 = E + jF$, what is the real part E ? [2] |
| 41 | (b) If $\mathbf{I}_2 = E + jF$, what is the imaginary part F . [2] |

Questions 42 to 44 [6]

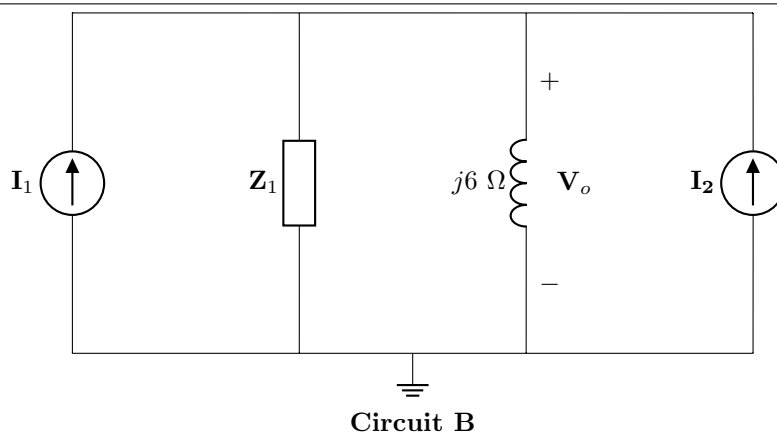
In **Circuit A**, when $R = 0 \Omega$, find the exclusive contribution, $\mathbf{V}_o^1$ from the voltage source to $\mathbf{V}_o$. The values x and y are given in ClickUP.



42 (a) If $\mathbf{V}_o^1 = G + jH$, what is the real part G ? [2]

42 (b) If $\mathbf{V}_o^1 = G + jH$, what is the imaginary part H ? [2]

When $R = 10 \Omega$, the Circuit A is transformed to **Circuit B** using source transformation. Determine $\mathbf{I}_1$ and $\mathbf{Z}_1$.



43 (a) If $\mathbf{I}_1 = K + jL$, what is the real part K ? [2]

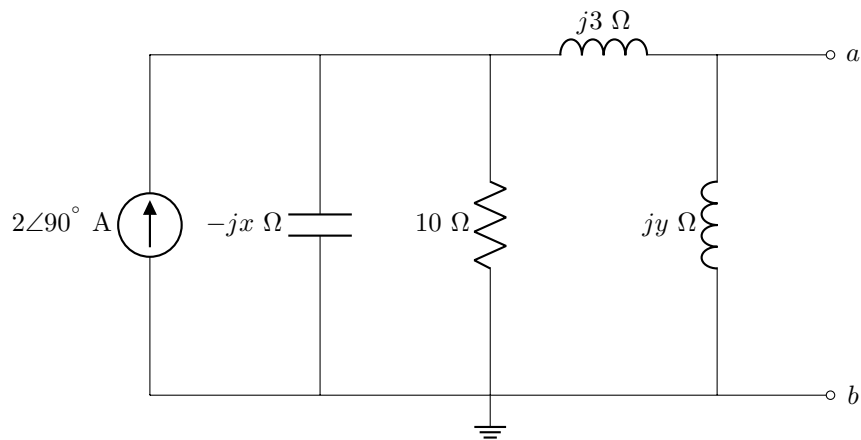
43 (b) If $\mathbf{I}_1 = K + jL$, what is the imaginary part L ? [2]

44 (a) If $\mathbf{Z}_1 = P + jQ$, what is the real part P ? [2]

44 (b) If $\mathbf{Z}_1 = P + jQ$, what is the imaginary part Q ? [2]

Questions 45 to 46 [4]

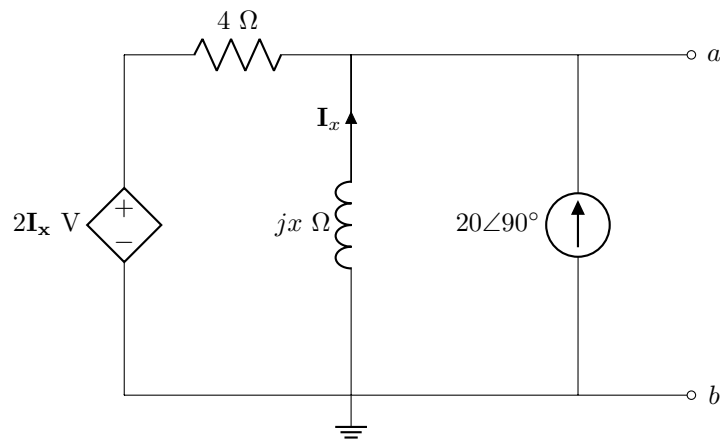
In the circuit below, find the Norton equivalent current $\mathbf{I_N}$ and impedance $\mathbf{Z_N}$ at terminals a-b. The values x and y are given in ClickUP.



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|----|---|
| 45 | (a) If $\mathbf{I_N} = S + jT$, what is the real part S ? [2] |
| 45 | (b) If $\mathbf{I_N} = S + jT$, what is the imaginary part T ? [2] |
| 46 | (a) If $\mathbf{Z_N} = U + jV$, what is the real part U ? [2] |
| 46 | (b) If $\mathbf{Z_N} = U + jV$, what is the imaginary part V ? [2] |

Questions 47 to 48 [4]

In the circuit below, find the Thévenin equivalent voltage $\mathbf{V_{Th}}$ and impedance $\mathbf{Z_{Th}}$ at terminals a-b. The value of x is given in ClickUP.



47	(a) If $\mathbf{V_{Th}} = X_1 + jY_1$, what is the real part X_1 ? [2]
47	(b) If $\mathbf{V_{Th}} = X_1 + jY_1$, what is the imaginary part Y_1 ? [2]
48	(a) If $\mathbf{Z_{Th}} = X_2 + jY_2$, what is the real part X_2 ? [2]
48	(b) If $\mathbf{Z_{Th}} = X_2 + jY_2$, what is the imaginary part Y_2 ? [2]

Total Section 3 / Totaal Afdeling 3: [32]

Grand Total: [96]